

Design of a Tendon Driven Underactuated Robotic Gripper for Acquisition of Non Cooperative Objects

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Abstract—Underactuated robotic grippers present a novel solution for the manipulation of non-cooperative objects, particularly in the unstructured environment of space. Where rigid grippers fail due to reliance on precise positional data and fixed geometry, compliant grippers are capable of passively adapting to irregular geometries, requiring only approximate pose information to achieve a successful grasp. Despite their potential, adaptive grippers for the capture of non-cooperative objects remains an underrepresented area in literature. This work proposes the design, simulation, and validation of an underactuated robotic gripper for the non-destructive acquisition of non-cooperative objects in space.

The design of this gripper follows a three-finger grasping modality, with spring-based flexure hinges that provide passive compliance. Underactuation of this gripper enables passive adaptation to irregular debris geometry, without fine-tuned actuator input. This approach facilitates successful grasping with no complex sensing or precise positioning.

In this paper we detail the development, simulation, and evaluation of the adaptive gripper, mounted to a seven degree of freedom robotic arm for testing. We detail the methodology behind the tendon-driven design and the integration of prototype parameters into simulation. Further, we provide qualitative and quantitative analysis of grasping success rate, simulation accuracy, and estimated contact forces during grasping trials. The underactuation reduces system complexity without sacrificing reliability, and presents a viable alternative for debris capture where object geometry and position are uncertain.

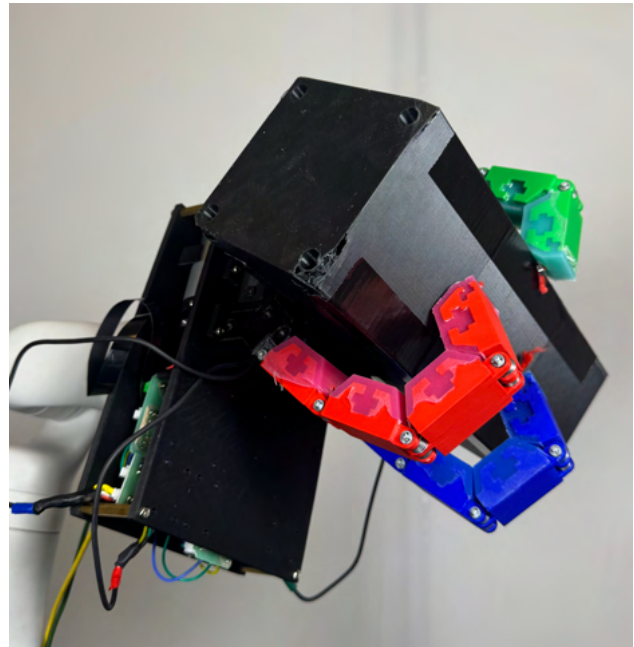


Figure 1: Gripper in the closed configuration with a captured cuboid.

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1. INTRODUCTION

The capability to safely acquire and manipulate non-cooperative objects is an important capability for future in-space operations. Tasks such as active debris removal, spacecraft inspection, on-orbit servicing, and retiring defunct systems require establishing a secure grasp and control over the target. Unlike cooperative targets that have standardized interfaces and grippable features, non-cooperative objects may present uncertain geometry, unknown orientation, and limited accessible grasp features. Further, relative motion and uncertainty in pose estimation can complicate path planning and execution of a reliable grasp. These challenges emphasize the need for an end effector that is capable of interacting with a wide range of object geometries without requiring precise knowledge of the target's properties.

Various active space debris removal (ADR) methods have been proposed, but are limited in operation. Tether-net systems display strong adaptability to various targets and fault tolerance, however, these systems rely on passive wrapping systems, which restricts their size to capture objects outside of their designed size envelope. Further, standard nets lack the ability to manipulate captured objects for inspection and maintenance. Lastly, these systems suffer from complex deployment dynamics, introducing navigation and control challenges [1] [2].

Compliant or adaptive mechanisms have also been investigated for space applications as a method of reducing sensitivity to target pose variation and target geometry. The hybrid-compliant capture system presented in [3], for example, combines rigid and compliant materials for the capture of uncooperative debris on a small scale. Alternative approaches have explored electro-adhesive compliant structures, such as the Reactive Electro-Adhesive Capture Cloth Mechanism (REACCH) [11]. These systems establish that the use of underactuated systems for capture of space debris is an active area of research.

Underactuated robotic fingers have been designed for the task of capturing non-cooperative objects using rapid closure and impact resistant design [5] [4]. These designs demonstrate the potential advantages of adaptive mechanisms for scenarios with uncertain target interaction. However, the final grasp configuration can be influenced by object geometry, mechanical properties of the finger, and contact sequence.

Underactuated robotic grippers employ fewer actuators than mechanical degrees of freedom, allowing the final joint configuration to be constrained by physical interactions between the gripper and the grasped object. Tendon-driven underactuation can apply a common torque input to a series of multiple joints. When the finger encounters an object, its motion becomes constrained, and further retraction of the tendon works to establish contact between the remaining unconstrained contact surfaces. This behavior results in the passive conformity of the underactuated system to the available grasp geometry, increasing contact area and resulting in a better grasp [6] [7] [8] [10]. One major challenge outlined by [7] is the ejection phenomenon in which continuous grasp sequences in underactuated hands can result in squeezing an object out of the hand rather than securing it. This highlights the need for a feedback system which provides insight into imparted forces during a grasping trial.

This work presents the design and experimental evaluation of a three-finger tendon-driven underactuated robotic gripper (Figure 1) for the capture of objects with uncertain geometry. Each digit contains four compliant revolute joints driven by a single tendon, enabling a multitude of final finger configurations based on the interaction between the passive joints and the capture object geometry. The gripper incorporates grasping feedback in the form of one load cell integrated alongside each tendon, which measures tendon tension. The prototype was mounted to a seven-degree-of-freedom robotic manipulator and evaluated using a range of representative object geometries.

The primary contributions of this work are:

1. The design and implementation of a three-finger, tendon-driven underactuated gripper capable of passively adapting to diverse object geometries;
2. Experimental characterization of grasp formation across multiple object geometries, including grasp success, repeatability, tendon loading, and estimated contact-force distributions;
3. Evaluation of grasp robustness under externally applied disturbances, including the influence of grasp configuration and disturbance direction on object retention;
4. Comparison between experimentally observed finger configurations and a simplified tendon-driven simulation model, highlighting the challenges associated with predicting the

multiple stable configurations of a compliant underactuated mechanism; and

5. A qualitative demonstration of dynamic target capture using a moving target surrogate.

The following sections describe the gripper design and operating principles, followed by the experimental methodology and simulation approach. Results are then presented and discussed, with an emphasis on grasp configuration, pose uncertainty, loading behavior, disturbance tolerance, and testing limitations.

2. GRIPPER DESIGN

The gripper was designed around two primary objectives: passive adaptation to objects with uncertain geometry and the incorporation of a minimal sensing system capable of monitoring grasp loading. This first goal was primarily achieved through the implementation of underactuated digits, wherein a single digit is made of four joints, all driven through a common tendon by a single actuator. This modality allows the finger to passively rearrange its shape in response to reactive forces, creating passive compliance. Further, this approach allows the digits of the gripper to achieve complex configurations that would require an actuator assigned to each joint to recreate. The final configuration is determined by tendon loading, joint compliance, and external contact forces. When a section of the finger contacts an object and its motion becomes constrained, continued tendon actuation allows the remaining joints to continue flexing, conforming to the object geometry. Reducing the number of actuators decreases the system complexity, and can improve control simplicity, which is advantageous for remote robotic systems. The second goal of implementing minimal sensing to monitor grasp loading was achieved by incorporating three load cells, one per digit, to provide quantitative data to indicate the forces being exerted during grasping. These measurements provide feedback on the forces transmitted through the grasping mechanism, and can be used to monitor changes in loading during gripper operation. Although high-resolution tactile sensing can provide detailed information, increasing sensor density also increases sensing and integration complexity [9]. Limiting sensor integration to one per digit reduces system complexity, while still providing quantitative information relevant to grasp execution.

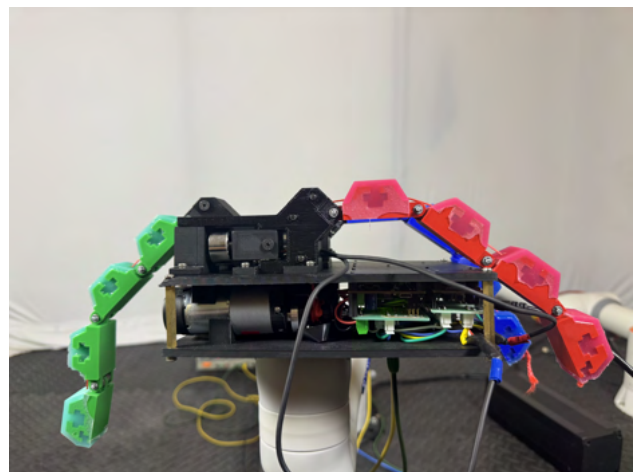


Figure 2: Gripper in the open configuration.

Mechanical Design

The gripper utilizes three identical underactuated digits arranged around a palm to provide a large capture envelope. The three finger design was selected to encompass smaller irregular objects, while still accommodating larger objects that only partially fit in the palm. Each digit consists of four contact surfaces and four mechanical revolute joints, allowing the digit to conform around encountered objects during closure. A CAD assembly of a finger is shown in Figure 3.

The fingers were designed with silicone molded contact surfaces to increase contact friction, as well as to distribute forces and prevent destructive grasps. The primary contact pad on each digit provides 807.4 mm^2 of available surface area.

Figure 2 shows the gripper in the relaxed, open position, with a side profile showing the location of load cells and actuators.

Tendon-Driven Underactuation

Each digit is actuated through a tendon-driven underactuated mechanism in which a single tendon provides the torque input required to flex the four joints. The tendon is routed along the length of the digit maintaining a 10 mm offset from the finger joint axes, and is anchored in the distal link. When tension is applied to the tendon, a moment is produced about each joint due to the offset between the joint and the tendon routing. The resulting joint moment can be approximated as

$$\tau_i = Tr_i \quad (1)$$

where τ_i is the moment about joint i , T is the applied tendon tension, and r_i is the 10mm offset. Tendons are each routed over a bearing in the base of the gripper, to redirect the tension along the axis of measurement of a load cell. Each tendon is terminated on a spool corresponding to its respective motor actuator. Each joint utilizes a torsional spring mechanism to return to its original open position, which opposes tendon tension. The equilibrium configuration of a finger that is not constrained by contact with an external object is determined by:

$$Tr_i = \tau_{s,i}(\theta_i) \quad (2)$$

where $\tau_{s,i}$ represents the restoring moment generated by the deflection of the torsional spring by the joint angle, θ_i . During closure, increasing tendon tensions initially cause the fingers to flex as the joints are unconstrained. Once one of the finger links makes contact with an object, the preceding joint becomes mechanically constrained and additional tendon tension is redistributed to the remaining unconstrained joints. This results in the digit reconfiguring in response to the encountered geometry. Because of this, the final grasp configuration and finger pose is not uniquely determined by the tension in the tendon, or the actuation of the motor. The resultant moment acting on a physically constrained joint becomes:

$$Tr_i = \tau_{s,i}(\theta_i) + M_{r_i} \quad (3)$$

where M_{r_i} is the reactive moment imparted on the i^{th} joint by the object being grasped.

An important consideration of the underactuated behavior is that a specified tendon tension does not correspond to a unique joint angle configuration. Due to the multitude of factors including initial configuration, friction, tendon routing, and joint compliance, the finger may settle into different stable configurations despite similar tendon loading. This characteristic, noted earlier, was observed during experimental testing and is discussed in the results section.

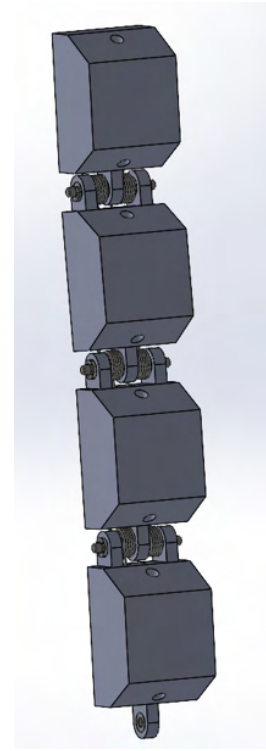


Figure 3: CAD Assembly of a single finger showing torsional springs and tendon routing holes

Design Materials

The gripper was constructed using a combination of rigid and compliant materials selected to provide structural integrity, reliable actuation, and compliant contact with captured objects. The base of the gripper consisted primarily of a 3D printed polylactic acid (PLA) housing to hold electronics and actuators. This allowed parts to be quickly manufactured and changed as necessary during the prototype stage. Rigid components were used for the palm and structural sections, while more compliant soft materials were used for contact surfaces. This provided structure for the gripper, while permitting the underactuated digits to grip the target.

The center palm and load cell housings were fabricated from Acrylonitrile Butadiene Styrene (ABS), due to its resilience under load and better impact absorption than PLA. This construction provided the rigidity to hold each load cell in a productive location, while avoiding damage from large loads during grasping.

Each finger was comprised of a PLA rigid section, and a silicone contact surface, cast from Dragonskin 30. PLA was selected due to its stiffness, ease of printing, and compatibility with the molding process. The compliant silicone contact surface provides deformation during contact with the target, increasing frictional grasp force while dispersing contact forces. Dragonskin silicone materials have also been used in compliant and soft robotic gripping applications [13]. Each digit incorporates four revolute joints that utilize torsion springs for restoring torque during tendon-driven flexion. The spring design produces a restoring force counter to the tendon tension, and provides resistance to flexion during grasp formation. The strength of these springs dictates the unconstrained behavior of the finger during actuation, which can be custom tuned by varying the strength of springs for individual joints. For the prototype reported in this paper, all springs share the same spring constant. Each revolute joint

was constructed using a hinge pin supported by bearings to provide repeatable isolated rotational motion. Several tendon materials were evaluated during prototype development, including braided fishing line, braided metal wire, Dyneema, and Vectran. Early materials exhibited either breakage or fraying during repeated high load operation. Ultimately, Vectran was selected for the final prototype due to its high tensile strength, resistance to creep, and use in other tendon-driven robotic systems [12]. All Vectran tendons remained intact throughout the extensive actuation presented in this work.

3. MODELING AND EXPERIMENTAL METHODOLOGY

Simulation Model

To characterize the relationship between tendon loading and finger configuration, a simplified simulation of the gripper was developed in MATLAB Simscape, with the mechanical model of the gripper shown in Figure 4. This model represents each four-link finger as a series of rigid links connected by revolute joints, with each revolute joint incorporating a spring constant based restoring torque. Each spring constant was tuned to match the properties of the torsion springs used in the physical model. The purpose of this model was to approximate the flexion behavior of the finger in the unconstrained free space environment.

Tendon tension served as an input to the simulation, and was converted to an equivalent moment applied to each joint. For a given tendon tension, the applied moment was calculated using the effective moment arm, as shown in equation 1. The per joint moment was input into the simulation, and the responsive joint angle was recorded. Physical joint limits were incorporated that correspond to the physical prototype, with the base joint limited to 40° and all other joints limited to 70° from the relaxed (straight) configuration.

Basic contact modeling between target geometries and contact pads were implemented. However, these contacts were intended to approximate the joint constraints caused by contact with an external object rather than to predict contact forces. This model does not explicitly capture all factors contributing to the final finger configuration, including detailed tendon routing, variation in joint friction, and deformation of contact surfaces.

Experimental Setup and Success Criteria

For testing and evaluation, the prototype was mounted to a seven DOF Kinova arm, providing a base for repeatable trials. For quantitative evaluation, the integrated load cells were used to record the tendon tension, and external force-sensitive-resistors (FSR) were placed on contact surfaces. This combination of sensors allowed for assessment of the quality of conducted grasps, as well as the effectiveness of the load-cell tension feedback monitoring. The gripper was extensively tested to quantify suitability in a variety of grasping trials meant to simulate possible encounters in the space environment. These experiments include: Grasping a variety of irregularly shaped non-uniform objects, capture of an object with pose uncertainty, repeatability testing, contact force estimation, disturbance rejection, and capture of a dynamic target.

Test Objects and Grasp Modes

A series of objects with varying characteristics were selected to evaluate the gripper's ability to accommodate objects with differing shapes, sizes, weights, and surface materials. Each

object was selected due to a unique property, to test a wide range of graspable features. The mass and general size of each object was recorded, and is reported in Table 1, along with the geometry classification of each object, and the category of grasp that was being tested. Further, Figure 5 provides a visual comparison of the different test objects.

Table 1: Tested Objects and Representative Grasp Features

Object	Mass (g)	Representative Grasp Feature
Strut	251.0	Cylindrical body
Circuit Board	362.0	Edge
Radiator	624.6	Frame/edge
Asteroid 1	133.4	Surface contour
Asteroid 2	82.9	Surface contour
Asteroid 3	141.2	Surface contour
Cuboid	594.5	Face/edge
Camera	1483.7	Housing/protrusion

Grasping Success Evaluation

The first experiment conducted was a success/failure evaluation of the gripper's performance on a subset of the test objects. Each object was suspended in the manipulation range of the test platform to simulate aspects of operation in a microgravity environment, and the gripper was tele-operated with the goal of capturing the target. The approach angle was kept consistent, while the orientation of each object was varied in between trials to introduce variation in the available grasp features. Each object was subjected to 10 grasping trials, with 5 different initial orientations. During each trial the gripper was not actuated until the object was visually verified to be within the capture envelope, to avoid unintentional operator caused failures.

A successful trial was considered when the gripper established contact, fully supported the object during lifting, and retained the object for 5 seconds during a 90 degree rotation of the object without manual intervention or regrasping. The success rate for each object was calculated as

$$S = \frac{N_{\text{successful}}}{N_{\text{total}}} * 100 \quad (4)$$

In addition to grasp success, qualitative observations of grasp configurations and finger-object contact were recorded.

Pose Uncertainty Evaluation

To evaluate the ability of the gripper to capture an object in the presence of pose error, grasping trials were conducted in which a positional or rotational offset was introduced, while the gripper orientation was kept the same. Each object was placed in a specified location, with the gripper following the same predefined path each trial. 40 trials were conducted each with Asteroid 1 and the Cuboid, with positional errors up to 15cm. Positional error distance was measured from the center of the palm, and was categorized into two types: positional error along the normal axis (perpendicular to the plane of the palm), and translation error (parallel to the plane of the palm). Orientation error consisted of a simple rotation of the object relative to the gripper.

The above two objects were selected to evaluate the gripper's ability to compensate for pose error on two objects of differing size, mass, and geometrical shape. Asteroid 1 represented an object with uncommon geometry that could be fully encompassed by the grasp envelope, while the Cuboid

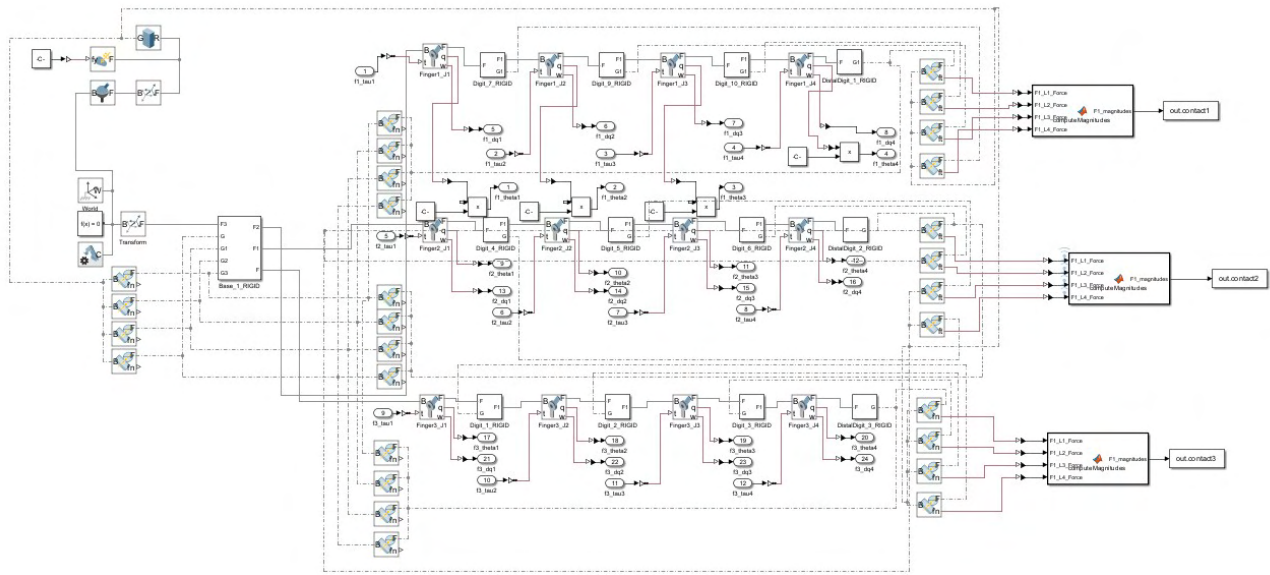


Figure 4: Simscape Gripper Mechanical Model

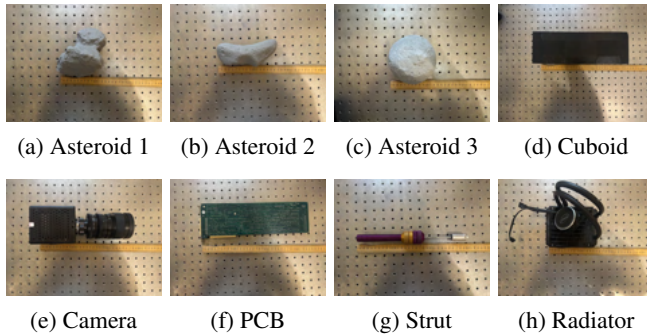


Figure 5: Test objects used throughout experiments.

served as a more conventional test object that the gripper could only grasp without fully encompassing.

Repeatability and Tendon Tension Measurement

Repeatability testing was conducted to evaluate the consistency of the gripper's grasping behavior under repeated trials. During these trials tendon tension was recorded to quantify variation in the mechanical loading of each grasp configuration. Because the underactuated finger can passively settle into different configurations depending on object geometry and contact conditions, the measured tendon tension provides additional data of grasp-to-grasp variability. Three objects: Asteroid 3, the cuboid, and the camera, were used as test objects due to their irregular, regular, and complex grasp geometries respectively. Each selected object was subjected to 10 trials where the gripper followed a preset approach path, and the object was placed at the same location relative to the gripper for each trial. The initial orientation of each object was maintained, and the gripper was commanded to actuate using the same command for each trial. Tendon tension was measured using the integrated load cell system during each grasp. Recordings were taken after a stable grasp was achieved, and the final tension reading was used. This data was compared against calibrated sensor recordings from a previous calibration and converted to calibrated tendon

tension (gf).

Contact Force Estimation

To estimate the distribution of contact forces during grasping, three force-sensitive-resistors (FSRs) were positioned along the contact surfaces of the three distal pads of a representative gripper finger, as shown in Figure 6. Measurements were taken on a single finger to provide an estimate of how contact loading was distributed along the final three contact pads. The first contact pad was ultimately neglected as it was observed that in many grasps this contact pad did not make contact with the target object, or made contact with the palm of the gripper during actuation.

Prior to grasping experiments, each FSR was calibrated using a known applied mass. The sensors were placed on a flat surface with an isolated finger contact pad placed on the sensor. Known masses were applied vertically to the contact pad, and the resulting sensor readings were recorded. After calibration recordings, a 4th order polynomial was fit to each sensor relating sensor reading to force.

During grasping trials, the FSR reading was recorded after a stable grasp was achieved. This FSR reading was converted to force using its corresponding calibration function. The resulting values represent the estimated local contact force under stable grasp conditions. These measurements were used to compare the relative distribution of contact loading between different grasp configurations across differing object geometries.

The FSR measurements should be interpreted as estimates of local contact loading rather than direct measurements of the total grasp force. Since the sensors were implemented into a compliant finger, finger flexion and deformation of the sensor introduced non-normal loading of the FSR substrate, resulting in changes in sensor output not due to normal contact force. Further, only one finger was equipped with sensors, and the results are not representative of the entire force distribution across the gripper. Regardless, the measurements provide quantitative data regarding the relative forces experienced during contact loading.

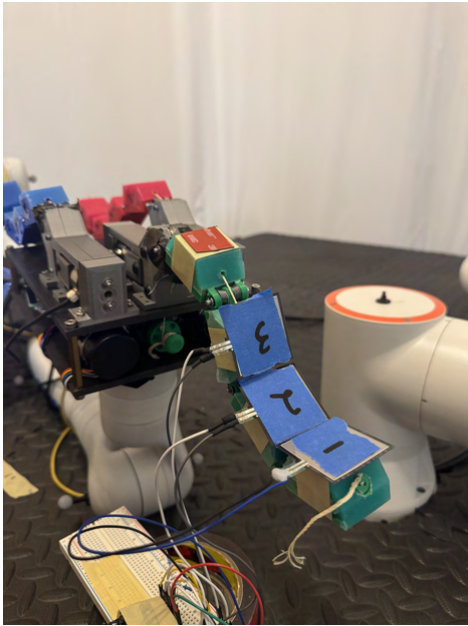


Figure 6: Placement of the FSRs on each finger contact pad.

Disturbance Rejection Evaluation

Disturbance rejection testing was conducted to evaluate the ability of the gripper to resist external forces applied to an already captured object. A cuboid was selected as the test object to provide a consistent grasp geometry. The cuboid was manually positioned in a stable grasp configuration and secured using the standard gripper closing procedure.

An external mass was attached to the grasped object using braided string, and released from a known vertical distance. This mass generated a disturbance as string became taut and transferred momentum to the grasped object. The magnitude of each disturbance was varied by changing the attached mass and release distance. Tested masses ranged from 200g to 2kg, while the release distances ranged from 5cm to 15cm.

The gravitational potential energy of each disturbance was calculated as

$$E = mgh \quad (5)$$

where m is the dropped mass, $g = 9.81m/s^2$ and h is the release height. This value was used to indicate the magnitude of each disturbance. This approach does not represent the exact energy or force transmitted to the object, as losses associated with the string, deformation, or motion were not directly measured.

Two disturbance vectors were evaluated. First, an optimal disturbance direction, where loading was applied in such a way that the geometry and mechanical structure of the gripper was best suited to resist the disturbance. This grasp orientation had two of the fingers underneath the object, so the disturbance was applied into these fingers. The second disturbance vector, referred to as the unfavorable or worst case disturbance direction, applied the load in a direction most likely to dislodge the object from the grasp. In this orientation the disturbance was applied perpendicular to the fingers, with no mechanical structure preventing slippage of the target. See Figure 7 for a visual of optimal and non-optimal grasping configurations.

Each mass/height combination was repeated 5 times, for a total of 50 trials. A trial was classified as successful if the object was retained by the gripper following the disturbance. Slippage or small movements of the object did not result in a

failure, and were not recorded.

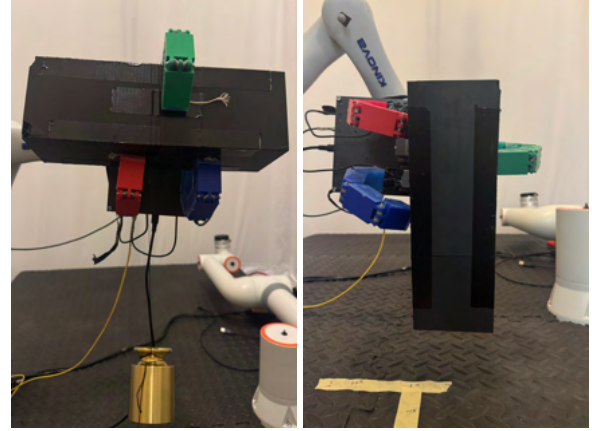


Figure 7: Disturbance Rejection Orientation showing Optimal (left) and Non-optimal (right) configurations

Dynamic Target Capture

To evaluate the gripper's ability to capture an object without precise prior knowledge of its position a dynamic target capture experiment was conducted. A small aerial drone was selected as a dynamic target due to its ability to reproduce controllable motion. During each trial, the drone was operated within the capture area of the gripper. Once the drone entered the capture envelope the gripper was commanded to close. The compliant, underactuated fingers then conformed to the available geometry of the drone.

Successful capture of the drone was defined as establishing a hold of the drone and retaining the target during arbitrary movement and rotation, without re-actuation. Quantitative data such as relative velocity, grasping forces, and capture probability were not recorded due to the unstructured nature of the experiment. Instead, these trials provide a qualitative demonstration that the gripper can acquire targets under relative motion, complementing the previously described controlled static grasping experiments. Further discussion and analysis are presented in the following section.

4. RESULTS AND DISCUSSION

Grasping Performance Across Object Geometries

Across the 70 grasping trials, the gripper achieved an overall successful capture rate of 85.7% (Table 2, Figure 8). Performance varied depending on test object geometry, demonstrating that the shape of the object and geometry available to the gripper influenced the grasp configuration. The highest success rates were observed for objects with large graspable features, including Asteroid 2, the Cuboid, and the Radiator. In contrast, the lowest success rates were observed for objects that had more difficult geometries, such as the strut, PCB, and camera.

Successful grasps were observed to occur more frequently when the target had structure that the fingers could gain purchase on, such as the knobby features of asteroid 2, or the thick texturing of the radiator. The size of the target also played a role in the likelihood of a successful capture, as objects smaller than the intended capture envelope, such as the strut, were difficult for the gripper to hold. In these trials, successful grasps often only had one finger holding the strut against the palm, while the other fingers never made contact

with the object. Despite this, these grasps were classified as successful when the object remained retained following closure. Another geometry that the gripper appeared to struggle with was the long flat geometry of the PCB. It was observed that this object would sometimes become lodged in the space between the fingers, resulting in a failed grasp. Overall, This behavior demonstrates that the underactuated mechanism was able to accommodate substantially different object geometries, without requiring a prescribed finger placement or uniform contact distribution. Further, the gripper demonstrated the ability to capture objects that fall outside of its original design, with a reasonably high success rate. The variation in success rate across geometries indicates that the outcome of a grasp is nevertheless dependent on the object's shape and accessible contact surfaces.

Table 2: Grasping Success Across Tested Object Geometries

Object	Trials	Successes	Success Rate (%)
Asteroid 1	10	8	80
Asteroid 2	10	9	90
Camera	10	8	80
Cuboid	10	9	90
PCB	10	8	80
Radiator	10	10	100
Strut	10	8	80
Overall	70	60	85.7

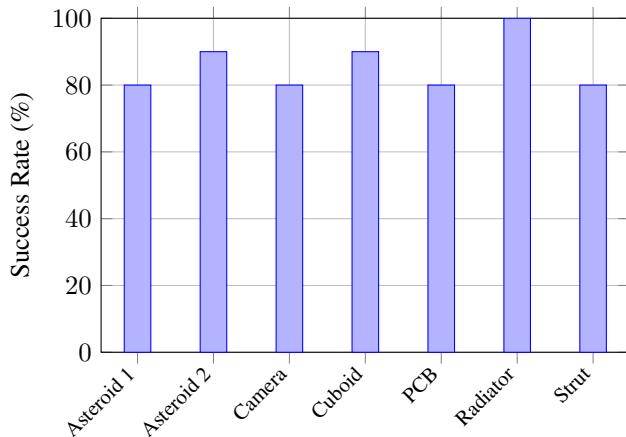


Figure 8: Grasp success rate for each tested object geometry.

Table 3: Grasp Success as a Function of Object Orientation

Orientation (°)	Trials	Successes	Success Rate (%)
0	14	12	85.7
45	14	13	92.9
90	14	12	85.7
135	14	13	92.9
180	14	10	71.4

Performance Under Pose Uncertainty

The pose uncertainty experiments revealed that the gripper demonstrated significant tolerance to both translational and rotational variations in target pose without requiring reorientation of the gripper prior to grasping. Tables 4 and 5 show that targets were still successfully captured with high

success rates for all objects subjected to a position error, and success rates remained consistent regardless of orientation error. It was noted that when actuated, the digits of the gripper sweep inwards towards the palm, creating a large capture error which proved especially beneficial for objects with larger normal positional errors. Despite being outside of the ideal capture envelope, the sweeping motion of the fingers was able to pull the object into a more optimal position where the other fingers could make contact and establish a grasp. This was seen across all objects with larger position errors. It is likely that if the fingers were constructed with additional links this sweeping motion could be utilized to create a much larger capture envelope.

It was observed from that the finger orientation varied greatly depending on the orientation of the object before capture, as shown in Table 3. For example during multiple trials involving the cuboid, irregular grasps were recorded in which the digits fully closed, and were making contact with dorsal side of the finger. Regardless, the gripper still showed resilience to changes in rotational orientation.

Repeatability and Tendon Loading

The tendon tension experiments demonstrated that the applied load was distributed unevenly among the three tendons depending on the resulting grasp configuration. While repeated grasps shared some similarities in loading characteristics, exact loads varied greatly. This is most likely primarily due to minor variations in the contact location in between trials. For objects of irregular geometry such as Asteroid 3, any variation in contact location could result in drastically different final grasp configurations, leading to differing tension readings. In contrast more symmetrical shapes such as the camera and cuboid have much more uniform contact surfaces. Slight variations in contact location likely do not have as large of an effect on the final grasp configuration, resulting in a lower coefficient of variation, as shown in Figure 9.

Despite this variation in tension readings, the experiment was conducted without a single failed trial, demonstrating a high level of repeatability in the presence of an assortment of grasping configurations.

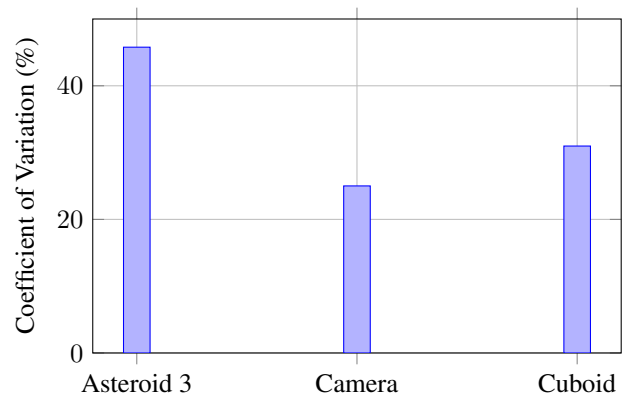


Figure 9: Variation in repeated final grasp tendon loading. Lower variation indicates greater loading repeatability.

Contact Force Distribution

The estimated contact forces displayed notable variation between trials conducted with differing object geometry. This disparity is to be expected from an underactuated system in which a single tension value does not correspond with a single

Table 4: Grasp Success Under Position Error

Position Error (cm)	Asteroid 1			Cuboid		
	Trials	Successes	Rate (%)	Trials	Successes	Rate (%)
0	5	5	100	5	5	100
5	5	4	80	5	5	100
10	5	5	100	5	3	60
15	5	4	80	–	–	–
20	–	–	–	5	4	80

Table 5: Grasp Success Under Orientation Error (20 trials for each object)

Orientation Error (°)	Asteroid 1 (%)	Cuboid (%)
0	100	100
45	100	90
90	80	100
180	100	–
270	100	–

Table 6: Repeatability of Final Grasp Tendon Loading

Object	Trials	Mean (g)	Std. Dev. (g)	CV (%)
Asteroid 3	10	3671.60	1680.74	45.78
Camera	10	11730.44	2933.88	25.01
Cuboid	10	5112.02	1583.76	30.98

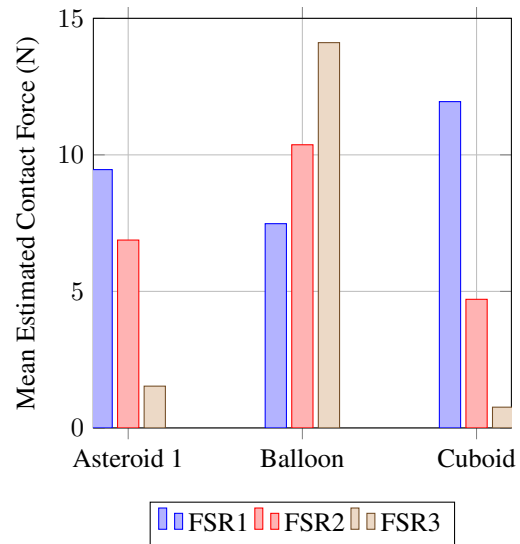
contact distribution. Instead, contact distribution for a single grasp configuration is determined by the interaction between tendon tension, finger configuration, object geometry, and finger pad contact sequence. This result indicates that the gripper is reaching different grasp configurations, but all recorded grasps in this set of experiments were successful. The availability of a multitude of force distributions leading to a successful grasp suggests that the gripper is not reliant on a single finger configuration for successful capture.

The estimated local contact force distribution is shown in table 6, and the contact force by FSR location is shown in Figure 10. By observing Figure 10, we can see that Asteroid 1 and the Cuboid share similar force distributions, with the contact pad on the distal contact pad recording the highest average force measurement across all trials. In contrast, the FSR placed on the proximal pad reported the lowest mean estimated contact force. This is due to the distal pad making contact first, which rearranges the geometry of the finger during contact and pushes the proximal pad away from the object, reducing the reading of this pad.

This was observed for rigid objects, however the trials involving the balloon demonstrate more consistent loading. This results from the balloons curved surface, which the compliant fingers were noted to have wrapped around, maintaining contact with all three contact pads. The main sensor loading remained within a relatively narrow range across repeated trials. This behavior is consistent with the intended design in which further tendon actuation allows additional sections of the compliant finger to conform to the object after initial contact.

Near-zero readings of the distal sensor (FSR3) should not necessarily be interpreted as an absence of interaction, as sensor bending was observed during finger flexion. Occasionally, when the preceding joint was near its maximum flexion, the FSR would be in contact with the surrounding joints, imparting a low reading. Because of this, these values are treated as approximate local contact estimates rather than

direct measurements of normal contact force.

**Figure 10:** Mean estimated contact-force distribution across sensing locations for representative object geometries. Asteroid trials affected by structural sensor contact were excluded.

Disturbance Rejection

Disturbance rejection testing demonstrated that an established grasp could withstand an external impulse applied in best and worst case orientations. Retention performance depended strongly on both the magnitude of the disturbance, as well as the direction of the load relative to the gripper structure.

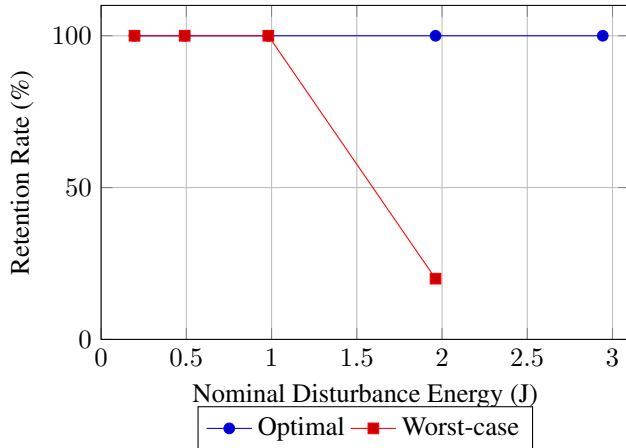
The overall disturbance rejection performance is shown in Table 8. In the optimal configuration, all disturbances tested were successfully resisted, with retention in all recorded trials. The placement of finger structures in the direction of the applied force allowed the gripper to successfully resist all external disturbance, maintaining control of the captured target. These disturbances acted primarily to drive the cuboid into the fingers holding it, allowing the disturbance to be opposed by the actuation of the tendon and finger geometry. In contrast, disturbances in the unfavorable direction resulted in translation towards an escape path where there were no physical structures to resist movement. Instead, these disturbances were only able to be resisted by the frictional contact forces imparted by the fingers, which were not measurable. Despite this, the gripper was still able to resist loads up to 2kg, with retention performance decreasing in trials with over 0.981 J. Consequently, the ability of the gripper to retain the object depended not only on disturbance magnitude but also direction of disturbance and the grasp configuration.

The dependence on disturbance direction demonstrates an im-

Table 7: Estimated Local Contact Force Distribution

Object	FSR1 (N)	FSR2 (N)	FSR3 (N)	Mean Pad Force (N)	Mean Tendon Tension (g)
Asteroid 1 ^a	9.46 ± 11.67	7.49 ± 1.28	11.95 ± 4.66	5.96 ± 4.65	7617.10 ± 4657.21
Balloon	6.88 ± 2.87	10.37 ± 2.8	4.71 ± 2.04	10.65 ± 2.41	11165.57 ± 2699.50
Cuboid	1.53 ± 0.60	14.11 ± 4.39	0.76 ± 0.56	5.81 ± 2.14	12819.43 ± 3876.93

portant characteristic of underactuated gripper design. Resistance to external disturbances is not uniform in all directions. Instead, disturbance tolerance is a result of finger compliance, object geometry, and grasp configuration. Disturbances that align with geometric constraints could be resisted more effectively than disturbances aligned with object escape paths.

**Figure 11:** Object retention as a function of nominal disturbance energy for favorable and unfavorable force vectors.

Simulation-to-Hardware Comparison

The Simscape model was evaluated by comparing simulated joint angles to experimentally measured joint angles over a range of applied tendon tensions in free space. The comparison was intended to determine whether the simplified tendon-to-joint torque model was able to reproduce the general kinematic behavior of the physical fingers, and to quantify the discrepancy between the simulation and prototype gripper.

Figure 12 shows the comparison between calibrated tendon tension, and both experimental and simulated joint angles. Across the tested loads, the simulation followed the general behavior displayed by the prototype finger, although the magnitude of individual joint rotations differed. This magnitude discrepancy was seen to decrease as the finger configuration approached joint limits, where both the physical and simulated joints were constrained.

The primary reason for the discrepancy between simulated and experimental joint angles - and why an underactuated system like this is difficult to simulate - is that tendon tension does not uniquely determine the configuration of the physical underactuated finger. For a given applied tension the experimental finger could settle into a multitude of stable configurations, while the simulation is deterministic. In practice, the final configuration of the finger is a result of initial conditions, friction, tendon routing, joint compliance, the sequence of contact locations, and mechanical interaction during flexion. Because of this, two experimental trials with identical tension loading could produce different joint-angles distributions. In contrast, simulated closures with identical tensions will result in identical simulated joint angle distributions.

This comparison indicates that the simulation captures the

general relationship between tension in the tendon and torques applied to the joints, but does not replicate the factors necessary to reproduce the exact finger flexion that was found experimentally. The simulation is appropriate for evaluating the qualitative kinematic behavior and approximate tendon tension required to achieve a certain closure rather than predicting final finger configuration or exact contact forces. Table 9 shows the comparison of simulation-to-hardware joint angles and error analysis.

Dynamic Target Capture

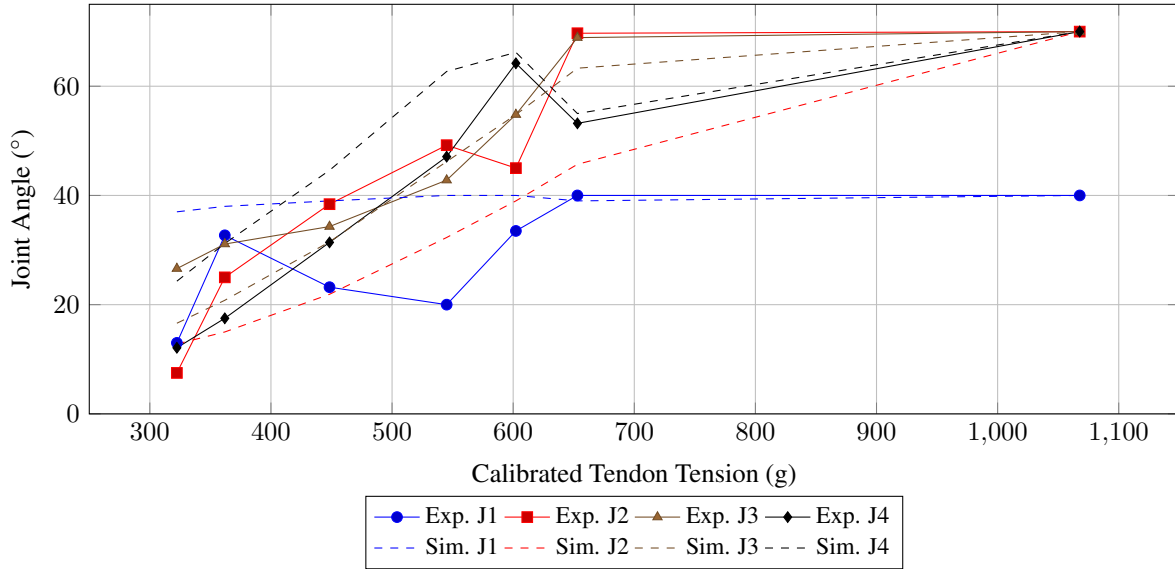
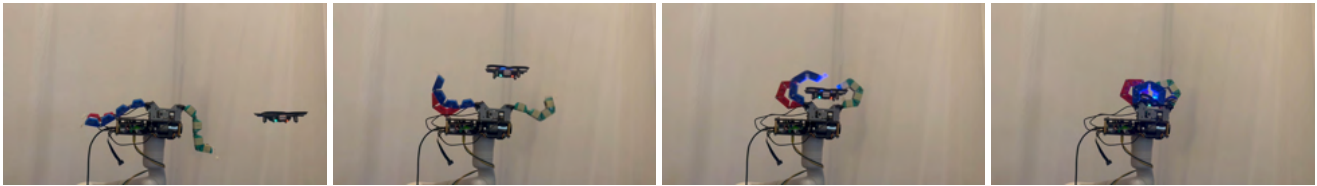
Dynamic target capture attempts demonstrated that the gripper was capable of successfully capturing a target that had uncertain motion relative to the gripper. Representative frames from a capture attempt are shown in Figure 13. During successful trials, the target entered the capture region without precise teleoperation of the gripper. Trials often involved a single finger making contact with the target, and the aforementioned sweeping motion pulling the target into the palm, where the other fingers could fully encompass it.

Unsuccessful capture attempts were noted to occur often when the drone moved unexpectedly at the point of contact. Instead of sweeping the drone into the capture area, occasionally the drone would be pushed away, resulting in a failed capture. These failures were primarily associated with operator error or instability of the drone movement rather than an inability of the fingers to conform to the target.

These trials do not provide a quantitative analysis of capture probability. Instead, they provide a proof-of-concept demonstration in which the compliant fingers, along with the closure mechanism of the gripper were able to establish a stable grasp on a moving target despite relative motion.

Table 8: Disturbance Rejection Performance

Force Vector	Mass (g)	Height (mm)	Nominal Energy (J)	Retained/Trials	Rate (%)
Optimal	200	100	0.196	5/5	100
Optimal	500	100	0.491	5/5	100
Optimal	1000	100	0.981	5/5	100
Optimal	2000	100	1.962	5/5	100
Optimal	2000	150	2.943	5/5	100
Worst-case	200	100	0.196	5/5	100
Worst-case	500	100	0.491	5/5	100
Worst-case	1000	100	0.981	5/5	100
Worst-case	2000	50	0.981	4/5	80
Worst-case	2000	100	1.962	1/5	20

**Figure 12:** Experimental and simulated joint angles as a function of calibrated tendon tension. The physical finger may reach multiple stable configurations under similar tendon loading, while the deterministic model predicts a single equilibrium configuration.**Figure 13:** Representative sequence of a successful dynamic target capture.**Table 9:** Simulation-to-Hardware Joint Angle Error

Joint	MAE (°)	RMSE (°)	Maximum Error (°)
Joint 1	10.37	13.61	24.00
Joint 2	11.24	13.62	24.00
Joint 3	4.59	6.06	10.30
Joint 4	8.36	10.43	15.60
Overall	8.64	11.36	24.00

Limitations and Relevance to Space Operations

All experimental testing was conducted under 1g, and consequently, do not reproduce the free-floating dynamics found with in-orbit operations. While some attempts were made

to replicate this environment by suspending objects during grasping trials, this does not replace in-depth testing in a more realistic environment. In particular, objects with flat geometries (such as the PCB) were found to be extremely difficult to pick up from a flat surface, where the object could not move freely into the capture envelope.

In a microgravity environment, a non-cooperative object would not be supported by any surface, allowing it to be maneuvered relative to the gripper to achieve a more optimal position. This removes the requirement to lift an object against gravity, and would allow the compliant fingers to conform to geometric features of the target more readily. The dynamic target capture experiment was evaluated qualitatively using a simple drone as a target. Since target velocity, relative position, and capture probability were not recorded,

these trials only demonstrate feasibility of capture under laboratory motion conditions rather than providing quantitative evaluation of performance.

Similarly, the disturbance rejection testing was used to characterize the gripper's ability to resist external forces applied to an already captured object along a specified vector. Since these tests were done in laboratory conditions under the effect of normal gravity, they do not reproduce the full dynamics expected during an orbital capture.

Contact forces were estimated using FSRs placed on a single finger, which is not indicative of the contact force distribution among the entire gripper. Further, sensor bending and contact between pads introduced uncertainty, and the measurements therefore represent local estimates of contact loading rather than a complete force distribution.

Despite the limitations of the experimental testing, the results demonstrate several capabilities that are relevant to non-cooperative space manipulation. First, the gripper was able to conform and capture geometrically diverse objects, including asymmetric, irregular, rigid/compliant, and complex structures. This is relevant to the capture of debris or scenarios where a target does not present a standard shape.

Second, the gripper successfully captured objects using a variety of grasping modalities. Stable grasps did not consistently require engagement from all three fingers, and retention could be achieved by leveraging the geometry of the target against the palm with minimal fingers. This behavior could be advantageous in the capture of non-cooperative targets, for which grasping features and contact geometry may be unknown before contact.

The results also support the use of mechanically compliant fingers to reduce the necessity of contact localization. The variation observed in contact force distribution and final grasp configuration shows that the fingers can redistribute contact in response to object geometry without requiring precise control of each joint.

Finally, the disturbance rejection results demonstrate that the compliance of the fingers can help to resist external motion in established grasps. This resilience is dependent on the direction of loading, as well as the magnitude of the external disturbance, but can be resisted by the geometry of the gripper. This is particularly relevant to free-floating targets, where post-contact motion may generate disturbances that must be resisted to maintain successful manipulation.

Future work should focus on evaluating the gripper under conditions that are more representative of the intended environment, including free-floating target motion, and controlled target motion, velocity, and rotation.

5. CONCLUSION

This paper presented a compliant, underactuated gripper for the capture and manipulation of non-cooperative objects with uncertain geometry and pose. The compliant finger structure allowed the gripper to conform to a variety of geometrically and structurally diverse objects, and the integrated tension sensing provided useful grasping feedback. Experimental evaluation demonstrated successful grasping across a variety of conditions, with an overall grasp success rate of 85.7% across 70 trials. Grasp and contact configurations were found to vary between trials, indicating that the gripper was able to successfully capture objects with a variety of finger pose and contact distributions.

Additional experiments demonstrated tolerance to variations in object pose and external disturbances, while repeatability trials showed the reliability of the system. Simulation displayed the ability to generally predict the response of a

finger in free space in response to a tendon tension. Finally, a qualitative dynamic target capture experiment further showed that the gripper could successfully capture, and retain, a target with unpredicted relative motion. These results collectively demonstrate the ability of the gripper to accommodate diverse geometry, variation in pose, contact configuration, and relative target motion. Despite several limitations of the experimental methodology, these experiments demonstrate the potential of compliant underactuated grasping as a suitable approach for capturing objects with uncertain geometry and pose in future space robotic systems.

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